

FlyRank ML Internship 2026

Content Underperformance Predictor — Refresh Opportunity Results PDF

118,092 Rows scored Anonymized BigQuery export	CatBoost Best model Selected by Precision@50	0.850 ROC AUC Ranking separation quality	0.820 Precision@50 Top-50 underperformance hit rate
16,806 High confidence Rows ready for manual review	0.385 Base rate (declining) Below-peer-median CTR prevalence	91.0 Top queue score 0–100 final priority score	client holdout Validation style GroupShuffleSplit by client_hash_id

How to read this quickly

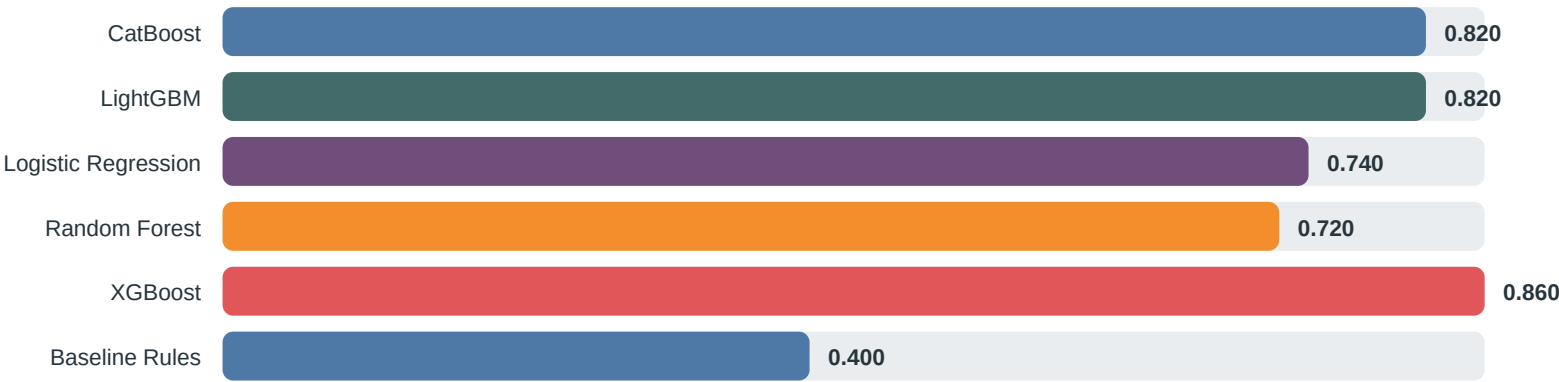
Question	Answer
What is the artifact?	A ranked queue of content pages predicted to have below-median CTR relative to their position bucket.
What is the model doing?	It predicts underperformance risk, then blends model probability with a transparent rule baseline for final prioritization.
What should mentors inspect first?	High-confidence rows with `refresh`, `refresh_and_review_ctr`, or `expand_and_refresh` actions.
What is not included?	No URLs, titles, keywords, domains, client names, or raw private identifiers.

Bottom line: this is a practical decision-support model for the internship capstone. It is not an automatic publishing decision and should be paired with editorial review. All claims are directional and measured, not causal.

1. Model Quality

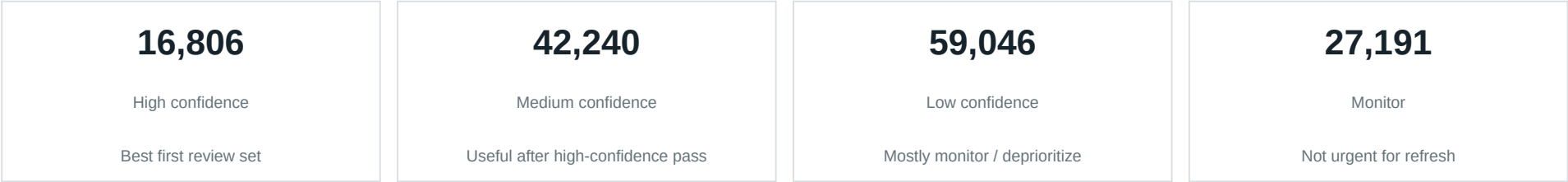
Model	ROC AUC	Avg precision	Precision@50	Recall	F1
CatBoost	0.850	0.666	0.820	0.902	0.705
LightGBM	0.850	0.663	0.820	0.915	0.694
Logistic Regression	0.783	0.548	0.740	0.879	0.672
Random Forest	0.825	0.601	0.720	0.911	0.680
XGBoost	0.848	0.660	0.860	0.909	0.700
Baseline Rules	—	—	0.400	—	—

Precision@50 comparison (model vs baseline)

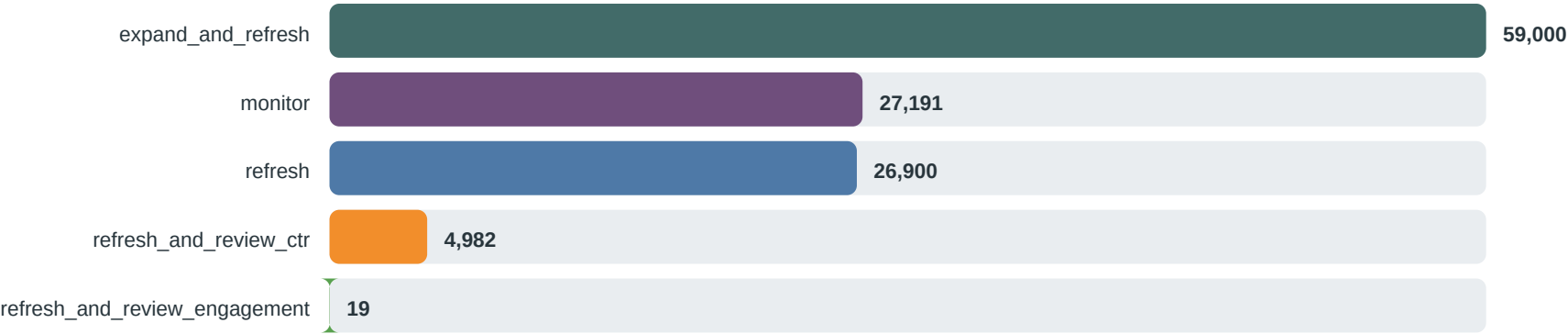


Metric	Plain-English interpretation
ROC AUC	How well the model ranks underperforming pages ahead of normal pages.
Average precision	Quality of the ranked list when positives are mixed with negatives.
Precision@50	How many of the top 50 recommended pages are actually below peer-median CTR.
Recall	How many underperforming pages are captured at the default 0.5 probability threshold.
Baseline Rules	Transparent rule-based score: old + never optimized + thin + no keyword.

2. Queue Shape



Suggested action mix

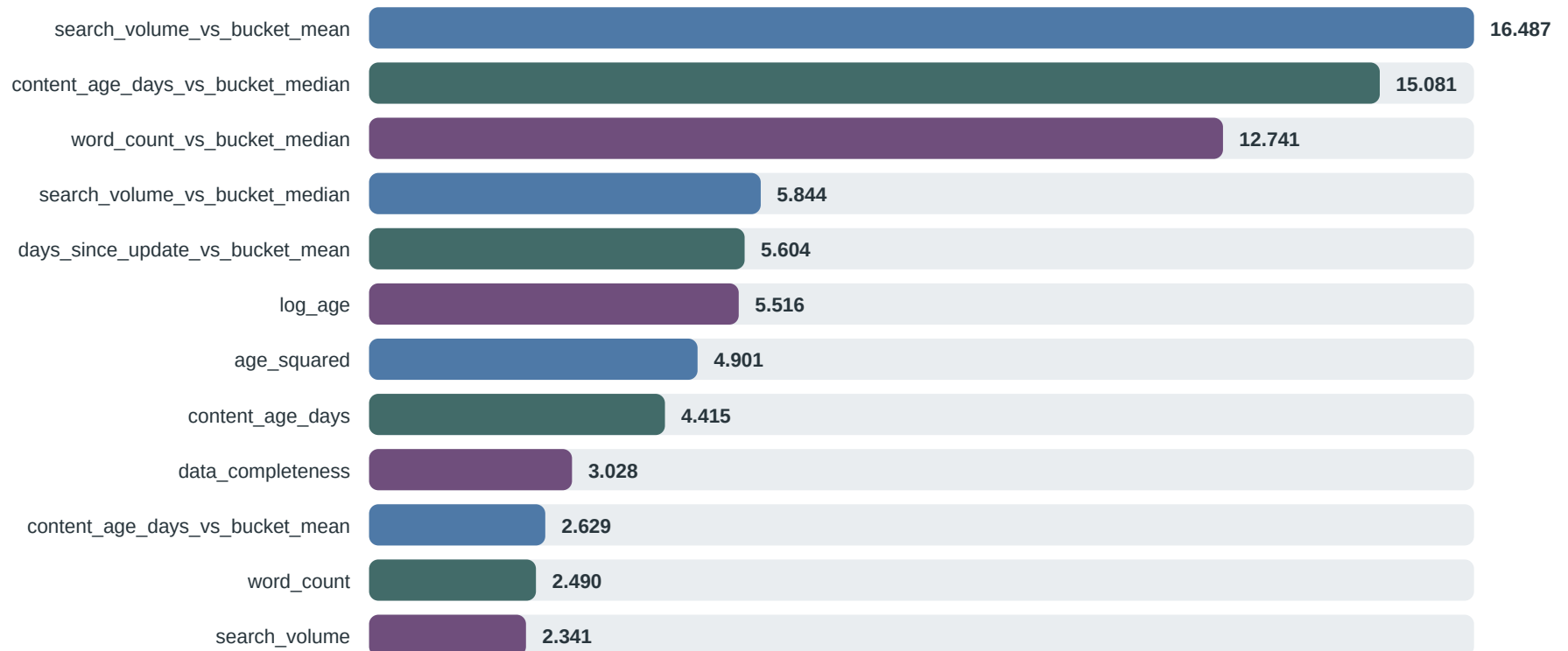


Confidence distribution



3. Why Pages Rank High

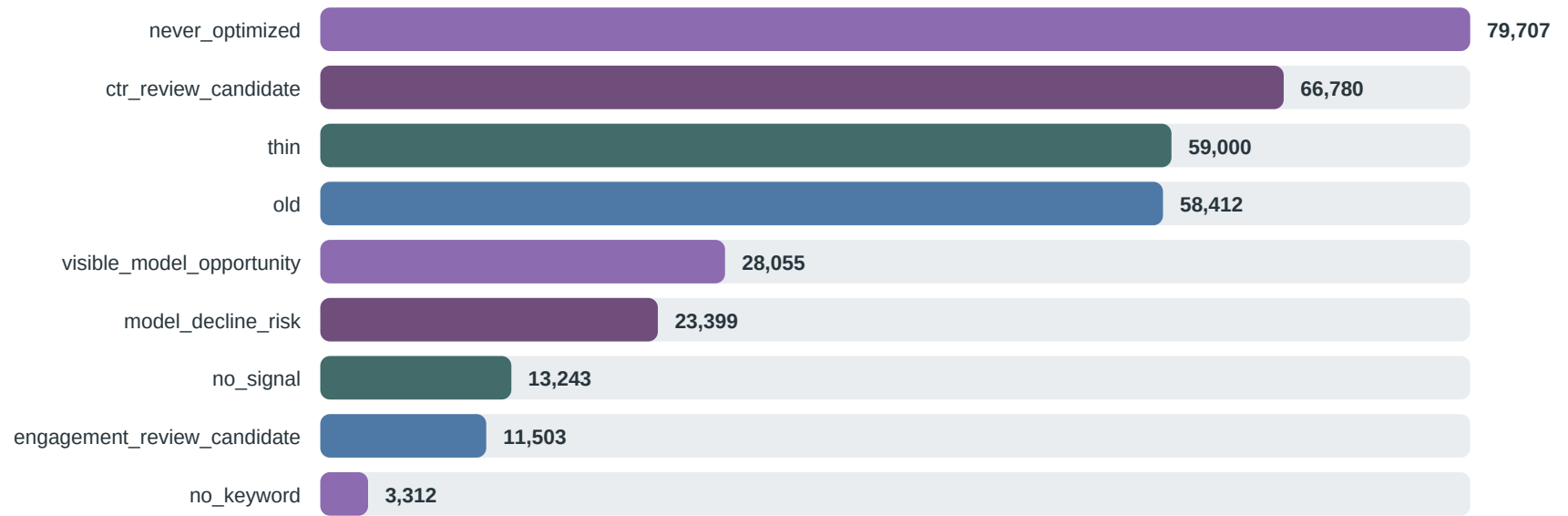
Top model features



Strong signals are mostly freshness and optimization related: `optimized_and_fresh`, `breadth_vs_depth`, `log_age`, visibility flags, and content depth. The model prioritizes pages that are old, thin, or never optimized.

4. Recommendation Reasons

Top recommendation reason codes



Reason codes explain why a row appears in the queue. They are designed for mentor review and intern explainability, not for automatic publishing decisions.

5. Top Queue Preview

These rows are anonymized. Use `content\_hash\_id` only to join back internally when mentors are ready to inspect real pages.

Rank	Score	Prob.	Confidence	Action	Impr.	Sess.	Declining	Reasons
1	91.0	0.871	high	expand + refresh	102	0	Yes	old, never optimized, thin, no keyword, model decline risk
2	90.6	0.973	high	refresh	101	0	Yes	old, never optimized, no keyword, model decline risk
3	90.3	0.969	high	refresh	157	0	Yes	old, never optimized, no keyword, model decline risk
4	90.3	0.969	high	refresh	135	0	Yes	old, never optimized, no keyword, model decline risk
5	90.2	0.860	high	expand + refresh	137	0	Yes	old, never optimized, thin, no keyword, model decline risk
6	90.1	0.858	high	expand + refresh	2,002	2	Yes	old, never optimized, thin, no keyword, model decline risk, visible model opportunity, ctr review candidate
7	89.5	0.957	high	expand + refresh	1,223	3	Yes	old, never optimized, thin, model decline risk, visible model opportunity, ctr review candidate
8	89.2	0.846	high	expand + refresh	106	0	Yes	old, never optimized, thin, no keyword, model decline risk
9	89.2	0.952	high	expand + refresh	1,377	0	Yes	old, never optimized, thin, model decline risk, visible model opportunity, ctr review candidate
10	89.0	0.843	high	expand + refresh	501	0	Yes	old, never optimized, thin, no keyword, model decline risk, visible model opportunity, ctr review candidate

Recommended next validation pass

Step	Validation action
1	Manually inspect the top 25 high-confidence rows in `refresh_queue.csv`.
2	Check whether the recommended action matches editorial context and business priority.
3	If accepted, convert this into the intern capstone task: reproduce, evaluate, explain, and improve the queue.
4	Avoid treating model score as an automatic publishing decision.